Blockchain Based Academic Certificate Verification System

A PROJECT REPORT

for

Major Project-II (CA401P)

Session (2025-2026)

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Submitted in partial fulfilment of the Requirements for the Degree of

MASTER OF COMPUTER APPLICATION

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Submitted to

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MAY, 2026

CERTIFICATE

Certified that **Khushi Jain (202410116100100) and Khushi Bora (202410116100099)** have carried out the project work having “**Blockchain Based Academic Certificate Verification System**” (**Major-Project-II, CA401P)** for **Master of Computer Application** from KIET Group of Institutions (an Autonomous college) under Dr. A.P.J. Abdul Kalam Technical University (AKTU), Lucknow under my supervision. The project report embodies original work, and studies are carried out by the student himself/herself and the contents of the project report do not form the basis for the award of any other degree to the candidate or to anybody else from this or any other University/Institution.

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge.

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Blockchain Based Academic Certificate Verification System

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ABSTRACT

Academic certificate fraud remains one of the most pressing challenges in educational institutions and hiring ecosystems, undermining trust in credentials and enabling misrepresentation of qualifications. Traditional paper-based and centralized digital verification methods are vulnerable to forgery, data tampering, and administrative inefficiencies that compromise the integrity of academic records. This report explores the application of blockchain technology for a decentralized, tamper-proof academic certificate verification system, offering a transparent and robust approach to authenticating educational credentials.

The study investigates the use of blockchain frameworks including Ethereum and Hyperledger Fabric, employing smart contracts to issue, store, and verify academic certificates without reliance on a central authority. Key components examined include certificate hashing, cryptographic digital signatures, distributed ledger storage, and QR-code-based instant verification mechanisms. The report evaluates different consensus mechanisms and compares the performance of permissioned versus permissionless blockchain architectures using metrics such as transaction throughput, verification latency, storage efficiency, and system scalability.

Implementation challenges including scalability constraints, key management, interoperability between institutions, and user adoption barriers are discussed along with proposed solutions. The research demonstrates that blockchain-based approaches can achieve high levels of data integrity and verification accuracy while significantly reducing reliance on manual authentication processes, outperforming traditional centralized credential management systems.

The findings indicate that smart contract automation and decentralized identity frameworks show strong promise in eliminating certificate forgery and streamlining multi-institutional verification workflows. This report contributes to the ongoing efforts to develop secure, transparent, and automated systems capable of protecting academic institutions and employers from credential fraud, ultimately enhancing trust and efficiency in the global educational landscape.

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ACKNOWLEDGEMENT

Success in life is never attained single-handedly. I am deeply grateful to my project supervisor, **Dr. Prashant Agrawal**, for his invaluable guidance, unwavering support, and encouragement throughout my project work. His enlightening ideas, constructive comments, and thoughtful suggestions have greatly contributed to the completion of this project.

I would also like to extend my heartfelt thanks to **Dr. Sachin Malhotra**, Professor and Dean, Department of Computer Applications, for his insightful feedback and administrative support on various occasions, which proved to be immensely helpful during critical stages of the project.

I am fortunate to have many understanding friends who have supported me in numerous ways during challenging moments. Their assistance and companionship have been a constant source of motivation.

Finally, my sincere gratitude goes to my family members and all those who have directly or indirectly provided me with moral support, encouragement, and assistance. Their unwavering belief in me and their continuous efforts to keep my life filled with happiness and joy made the completion of this project possible.

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INTRODUCTION

1.1 GENERAL

In the contemporary digital era, academic credentials have become fundamental assets for individuals seeking educational advancement and professional opportunities, with millions of certificates and degrees issued annually across institutions worldwide. However, this growing reliance on formal qualifications has made academic certification increasingly vulnerable to fraud, with certificate forgery and credential misrepresentation emerging as particularly damaging threats to educational integrity. Certificate fraud involves deliberate attempts to fabricate, alter, or misuse academic qualifications such as degrees, diplomas, and transcripts by individuals seeking to misrepresent their educational background. According to recent studies, a significant percentage of job applicants misrepresent their academic credentials, resulting in financial losses for organizations, compromised institutional reputations, and erosion of trust in the global education system.

Traditional academic verification methods, which rely on manual processes, centralized databases, and paper-based documentation, have proven inadequate against the evolving sophistication of modern credential fraud. Fraudsters continuously exploit vulnerabilities in legacy systems through document forgery, unauthorized record manipulation, and impersonation of legitimate awarding bodies, bypassing conventional verification defences. This escalating threat landscape necessitates the development of decentralized systems capable of issuing, storing, and verifying academic credentials in a transparent and tamper-proof manner. Blockchain technology, with its immutable distributed ledger and cryptographic security, offers a transformative solution to this critical challenge in academic credential management.

1.2 OVERVIEW OF INTELLIGENT EMAIL PHISHING DETECTION

This project focuses on developing a secure blockchain-based academic certificate verification system leveraging decentralized ledger technology to issue, store, and authenticate educational credentials with high integrity and transparency. The system employs a comprehensive approach that includes data collection from institutional certificate records, feature structuring from various credential components (student identity, course details, issuing authority, and timestamps), and the implementation of smart contract mechanisms for automated certificate validation.

The project encompasses the entire blockchain development pipeline, from system architecture design and smart contract development to deployment, testing, and real-world integration considerations. Various blockchain frameworks including Ethereum, Hyperledger Fabric, and other distributed ledger technologies are explored and compared based on their performance, scalability, and suitability for academic use cases. The system is designed to extract and encode meaningful credential attributes such as institutional digital signatures, cryptographic hash values, certificate metadata, ownership verification protocols, and immutable audit trails that distinguish authentic academic qualifications from fraudulent or tampered documentation.

1.3 OBJECTIVES OF THE SYSTEM

The primary objectives of this Blockchain Based Academic Certificate Verification system are:

- Develop a robust blockchain-based certification model capable of accurately issuing and authenticating academic credentials with minimal risk of forgery, tampering, or unauthorized modification.
- Identify and encode the most relevant attributes from academic credential data that effectively characterize authentic certificates, including institutional details, metadata, cryptographic signatures, and student information.
- Compare and evaluate multiple blockchain frameworks and consensus mechanisms to determine the most effective approach for certificate verification in terms of security, scalability, transparency, transaction speed, and computational efficiency.
- Create a scalable decentralized system architecture that can process verification requests in real-time and adapt to emerging credential fraud techniques through continuous smart contract upgrades and protocol improvements.

1.4 PROBLEM STATEMENT

Despite the critical importance of academic integrity, institutions and employers continue to face increasingly sophisticated credential fraud challenges. The current problems in academic certificate verification include:

- **Widespread Certificate Forgery:** Fraudsters continuously refine their techniques, producing counterfeit academic documents that closely replicate authentic certificates through advanced printing and digital manipulation, making them difficult to distinguish using traditional verification methods.
- **Slow Verification Processes:** Traditional verification systems often involve lengthy manual authentication procedures, leading to significant delays in credential confirmation, which disrupts recruitment operations and reduces confidence in institutional verification frameworks.
- **Lack of Centralized Trust:** New fraudulent certificates that exploit unrecognized institutional formats or impersonate lesser-known awarding bodies cannot be detected by conventional systems that rely on limited institutional databases.
- **Absence of Interoperability:** Siloed institutional record systems require manual cross-referencing and cannot automatically validate credentials across different organizations, creating persistent gaps between certificate issuance and reliable verification.
- **Resource Constraints:** Many existing verification solutions are administratively expensive or require extensive manual intervention, making them impractical for real-time deployment in high-volume recruitment and academic admission environments.

This project addresses these challenges by developing a decentralized, tamper-proof system that can authenticate academic certificates based on cryptographic validation rather than manual inspection, providing both high verification accuracy and operational efficiency while adapting to emerging credential fraud techniques.

1.5 TARGET AUDIENCE

This project is designed to benefit multiple stakeholders in the academic credentialing and professional verification ecosystem:

- **Universities and Educational Institutions:** Academic bodies of all sizes seeking to protect the integrity of their issued credentials, prevent certificate forgery, and streamline their official document authentication processes.
- **Employers and Recruitment Agencies:** Corporate organizations and hiring platforms looking to enhance their candidate screening processes with advanced credential authentication and instant verification capabilities.

- **Government and Regulatory Bodies:** Public sector authorities and accreditation agencies that require effective tools for monitoring and validating academic qualifications across national and international educational frameworks.
- **Financial and Professional Licensing Institutions:** Banks, certification bodies, and professional associations that frequently require verified academic credentials to authorize access to specialized roles and regulated financial services.
- **Healthcare Organizations:** Medical institutions and licensing boards that must verify practitioner qualifications and comply with regulatory standards while defending against fraudulent credential submissions.
- **Individual Certificate Holders:** Graduates and students who desire secure, portable, and instantly shareable digital credentials that reliably demonstrate their authentic academic achievements to prospective employers.
- **Researchers and Academics:** Blockchain researchers and students interested in distributed ledger applications for credential management, digital identity verification, and academic record security.

1.6 PROJECT SIGNIFICANCE

The significance of this blockchain-based academic certificate verification system extends across technical, institutional, and societal dimensions:

- **Enhanced Credential Integrity:** By providing accurate, real-time certificate authentication, the system significantly reduces the risk of successful credential fraud, protecting institutional reputations and preventing qualification misrepresentation across professional environments.
- **Proactive Fraud Prevention:** Unlike reactive verification approaches, blockchain technology enables proactive identification of tampered or fraudulent certificates, allowing institutions and employers to stay ahead of evolving document forgery techniques.
- **Cost Reduction:** Automated verification reduces the need for extensive manual credential review and administrative follow-up, lowering operational costs associated with traditional academic record authentication and management.
- **Institutional Continuity:** By minimizing verification errors and delays, the system ensures that legitimate credential authentication proceeds uninterrupted, maintaining productivity and operational efficiency across recruitment and admissions workflows.
- **Compliance and Regulatory Requirements:** The system helps institutions meet academic accreditation standards and data protection regulations by implementing advanced cryptographic verification and transparent audit trail mechanisms.
- **Scalability and Adaptability:** Blockchain-based approaches can scale to handle large volumes of certificate requests and continuously accommodate new

institutions, making them suitable for adoption across educational ecosystems of varying sizes.

- **Contribution to Cybersecurity Research:** This project contributes to the growing body of knowledge on applying distributed ledger technology to academic credential management challenges, potentially inspiring future innovation in decentralized identity verification systems.

1.7 LIMITATIONS OF THE SYSTEM

While this blockchain-based academic certificate verification system offers significant advantages, several limitations must be acknowledged:

- **Data Dependency:** The system's performance is heavily dependent on the quality, completeness, and accuracy of certificate data recorded on the blockchain. Incomplete or incorrectly entered institutional records may result in suboptimal verification capabilities for certain credential types.
- **Smart Contract Vulnerabilities:** Sophisticated attackers may exploit coding vulnerabilities within deployed smart contracts to manipulate certificate records, potentially compromising the integrity of the verification process if contracts are not rigorously audited.
- **Implementation Complexity:** Effective blockchain-based verification requires careful system design and development. The system's performance may degrade if critical architectural decisions are overlooked or if smart contract logic is improperly implemented.
- **Computational Requirements:** Advanced blockchain frameworks, particularly those employing complex consensus mechanisms, may require significant computational resources for transaction processing and network maintenance, potentially limiting deployment in resource-constrained institutional environments.
- **Real-Time Processing Challenges:** Achieving low-latency verification in high-volume credentialing environments presents technical challenges, particularly when implementing computationally intensive consensus protocols across distributed nodes.
- **Scalability Constraints:** No blockchain system achieves unlimited scalability. High volumes of simultaneous certificate verification requests may cause network congestion, while storage demands for large credential datasets may strain distributed infrastructure.
- **Multilingual and Cultural Context:** The system may face challenges in verifying academic credentials issued by institutions in different countries, or those following region-specific formatting and accreditation standards not represented in the system.

CHAPTER 2

FEASIBILITY STUDY

2.1 TECHNICAL FEASIBILITY

The implementation of a blockchain-based academic certificate verification system is technically feasible with current technologies. The project leverages open-source frameworks such as Ethereum, Hyperledger Fabric, and Truffle Suite, which provide robust implementations of distributed ledger mechanisms including smart contracts, consensus protocols, and cryptographic hashing algorithms. Solidity offers extensive support for smart contract development and deployment, while publicly available blockchain development tools like Ganache and Remix IDE provide adequate environments for testing and simulation of certificate issuance and verification workflows.

2.2 ECONOMIC FEASIBILITY

From an economic perspective, this project presents a favourable cost-benefit ratio. Development costs include personnel expenses and computational resources, which are modest compared to potential losses from phishing attacks. Organizations spend millions annually on breach recovery and reputational damage, making the detection system economically justified. Long-term operational costs remain minimal using cloud platforms, while open-source frameworks eliminate licensing fees.

2.3 Market Research

The From an economic perspective, this project presents a favourable cost-benefit ratio. Development costs include personnel expenses and infrastructure resources, which are modest compared to potential losses incurred from credential fraud and its consequences. Organizations and institutions spend considerable resources annually on manual verification procedures, legal disputes arising from fraudulent qualifications, and reputational damage, making the blockchain verification system economically justified. Long-term operational costs remain minimal using decentralized cloud-based node infrastructure, while open-source blockchain frameworks such as Hyperledger Fabric and Ethereum eliminate expensive proprietary licensing fees.

2.4 Existing Solutions

Current academic certificate verification solutions include traditional manual verification processes, centralized institutional databases, and paper-based authentication methods employed by universities and awarding bodies worldwide. Commercial products such as Digitary, Incredible, and Parchment offer digital credentialing services with some blockchain integration capabilities. However, these solutions often rely heavily on centralized record management and proprietary verification frameworks, requiring constant administrative oversight and institutional coordination. Open-source tools like Open Badges provide basic digital credential frameworks but lack sophisticated decentralized authentication and cryptographic immutability mechanisms.

2.5 Gap Analysis

Existing solutions exhibit several critical gaps that this project addresses. Traditional systems struggle with sophisticated certificate forgery techniques and suffer from slow manual verification processes, disrupting timely credential authentication for employers and academic institutions. Most commercial solutions are expensive and lack transparency in their verification mechanisms, making customization and cross-institutional integration considerably difficult. Current systems show limited interoperability across different educational frameworks and often require extensive manual administrative intervention for credential updates and record management. There is a significant need for a decentralized, cost-effective system that combines cryptographic security with smart contract automation, provides real-time verification with minimal authentication errors, and continuously adapts to emerging fraudulent credential techniques without constant manual institutional oversight.

CHAPTER 3

PROJECT OBJECTIVE

The main objective of the Blockchain-Based Academic Certificate Verification System is to develop a robust, efficient, and decentralized platform for issuing, managing, and authenticating academic credentials across educational institutions and professional organizations. The system aims to streamline the entire verification process—from certificate issuance and cryptographic encoding to identity validation, real-time authentication, and continuous smart contract improvement—while ensuring integrity, transparency, and minimal verification errors.

Implementing Advanced Blockchain Technology: The system will leverage multiple blockchain frameworks including Ethereum, Hyperledger Fabric, and distributed ledger protocols to create a comprehensive credential verification framework. The smart contracts will be deployed across decentralized networks, continuously updated with new institutional participants, and optimized for real-time performance to protect employers and institutions from evolving academic credential fraud techniques.

i. To decentralize academic credential management through an automated blockchain verification platform.

ii. To reduce manual certificate inspection and minimize human error in credential authentication.

iii. To automate certificate issuance and verification using smart contracts on distributed ledger networks.

iv. To maintain updated institutional records and adapt to emerging credential fraud techniques.

v. To enable rapid certificate authentication and instant response for detected fraudulent submissions.

CHAPTER 4

HARDWARE AND SOFTWARE REQUIREMENTS

4.1 Hardware Requirements

- A standard computer/laptop with minimum **2.0 GHz processor**
- **4 GB RAM** (minimum), **8 GB recommended**
- **500 GB storage** or more
- Stable internet connection
- Server machine (if deployed on local server)

4.2 Software Requirement

i. **Blockchain Development Frameworks**

- Solidity 0.8+
- Ethereum (for smart contract deployment)
- Web3.py (for blockchain interaction)
- Truffle Suite (for contract management)
- IPFS (for decentralized document storage)

ii. **Development Tools**

- Remix IDE (for smart contract development)
- VS Code / PyCharm (IDE)
- Git (version control)

CHAPTER 5

PROJECT FLOW

5.1 Development Methodology

The development of the Blockchain-Based Academic Certificate Verification System follows the Agile Software Development Methodology. Agile allows the system to be developed in small, manageable increments where each module is built, tested, and improved based on continuous feedback. This approach helps maintain flexibility, improves smart contract reliability through iterative refinement, and ensures better verification performance.

The development process begins with requirement analysis where certificate verification needs are identified, and institutional credential datasets are collected from university records and accreditation repositories. During the planning phase, the project is divided into smaller sprints focusing on specific tasks such as smart contract development, cryptographic hashing, certificate issuance, and verification workflows. System design involves creating the blockchain architecture, defining credential data structures, and designing the verification workflow with appropriate data flow diagrams.

In the development phase, certificate data cleaning and preprocessing are performed using Web3.py and Truffle Suite. Smart contract algorithms are implemented to encode and authenticate credential information including institutional details, student identity, and cryptographic signatures. Multiple blockchain frameworks including Ethereum and Hyperledger Fabric are deployed and evaluated for performance and scalability. The testing phase includes smart contract validation, performance testing using metrics like transaction speed, verification accuracy, and system scalability, and comparison of different consensus mechanisms to select the most suitable approach.

5.2 Data flow Diagram

LEVEL 0:

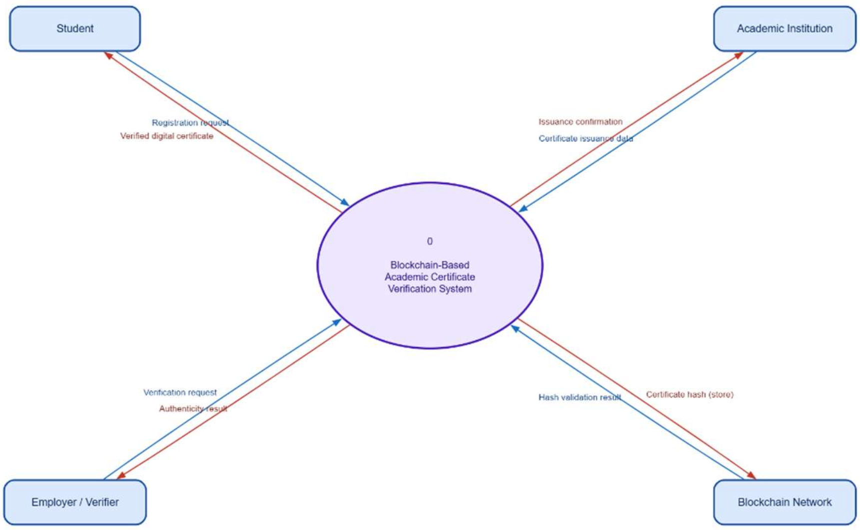

Figure-5.2.1

LEVEL 1:

The Level 1 DFD expands the central system into seven sub-processes across two flows:

Issuance flow (Part 1): The university submits certificate data → process 1.0 validates it → process 2.0 generates a cryptographic hash (stored in DS1) → process 3.0 deploys the smart contract to DS2 (blockchain ledger) → process 4.0 issues the digital certificate to the student. Validation errors are returned directly to the institution.

Verification & admin flow (Part 2): The employer submits a certificate ID or QR code → process 5.0 parses the request → process 6.0 queries DS1 for the hash record → process 7.0 validates the hash against DS2 (blockchain ledger) and returns the result to the employer. All events are logged to DS3, which feeds audit reports to the system administrator, who in turn manages configuration updates.

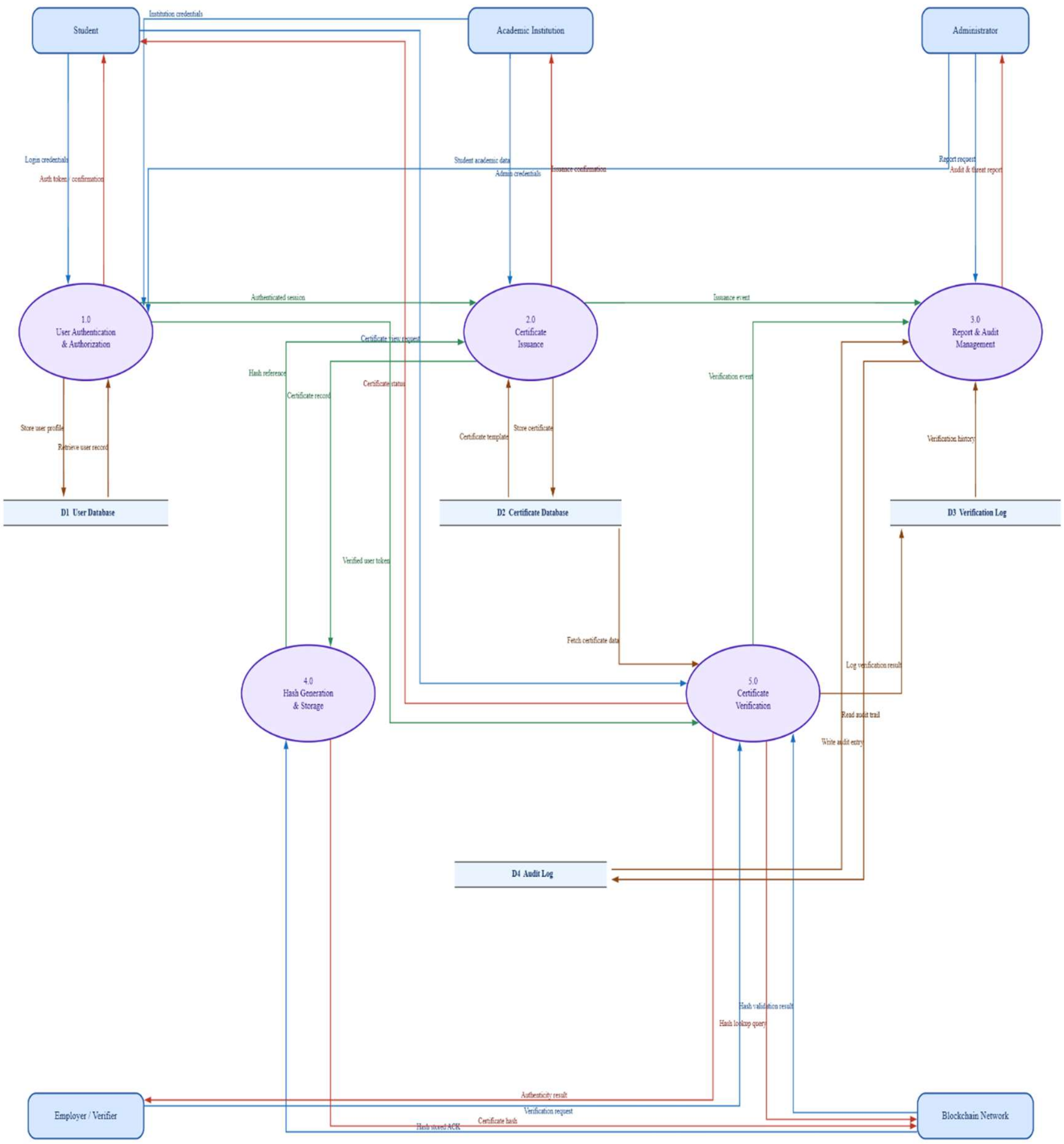

Figure-5.2.2

5.3 ER Diagram

The ER diagram includes 9 entities with the following relationships:

- **INSTITUTION** is the root entity — it enrolls students, offers courses, and issues certificates.
- **STUDENT** belongs to one institution and can hold multiple certificates.
- **COURSE** is offered by an institution; each certificate is tied to one specific course.

- **CERTIFICATE** is the central entity linking student, institution, and course, storing the cryptographic hash and IPFS content ID.
- **SMART_CONTRACT** has a strict one-to-one relationship with a certificate, recording the blockchain transaction address and network details.
- **VERIFICATION_LOG** captures every verification event, connecting a certificate to the verifier who checked it.
- **VERIFIER** (employer or recruiter) can perform many verifications across different certificates.
- **ADMIN** manages the system, and every admin action is recorded in AUDIT_LOG for traceability.

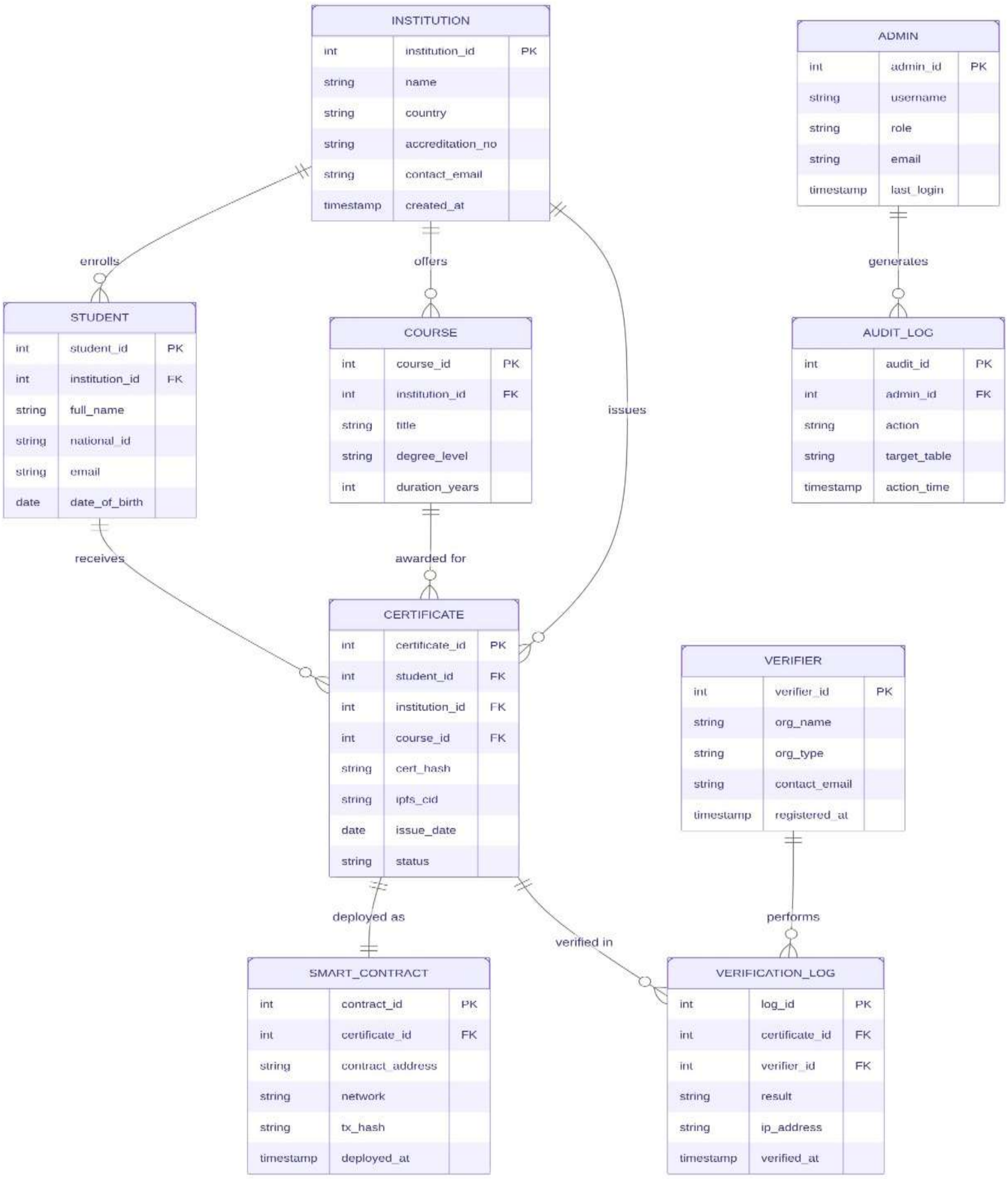

Figure-5.3.1

5.4 Use Case Diagram

The Use Case Diagram covers four actors and ten use cases:

Institution interacts with two use cases — registering on the platform and issuing certificates to students. Issuing a certificate automatically includes generating a cryptographic hash (which in turn includes deploying the smart contract), shown as «include» dependencies.

Student can view their issued digital certificates and share them with third parties such as employers or academic bodies.

Employer/Recruiter submits verification requests and views the verification result. The request verification use case includes the blockchain validation step, which in turn produces the viewable result.

System Administrator manages system configuration and reviews audit logs, covering governance and oversight responsibilities.

The teal-coloured ovals represent internal system use cases triggered automatically as part of other interactions, while purple ovals represent direct actor-initiated actions.

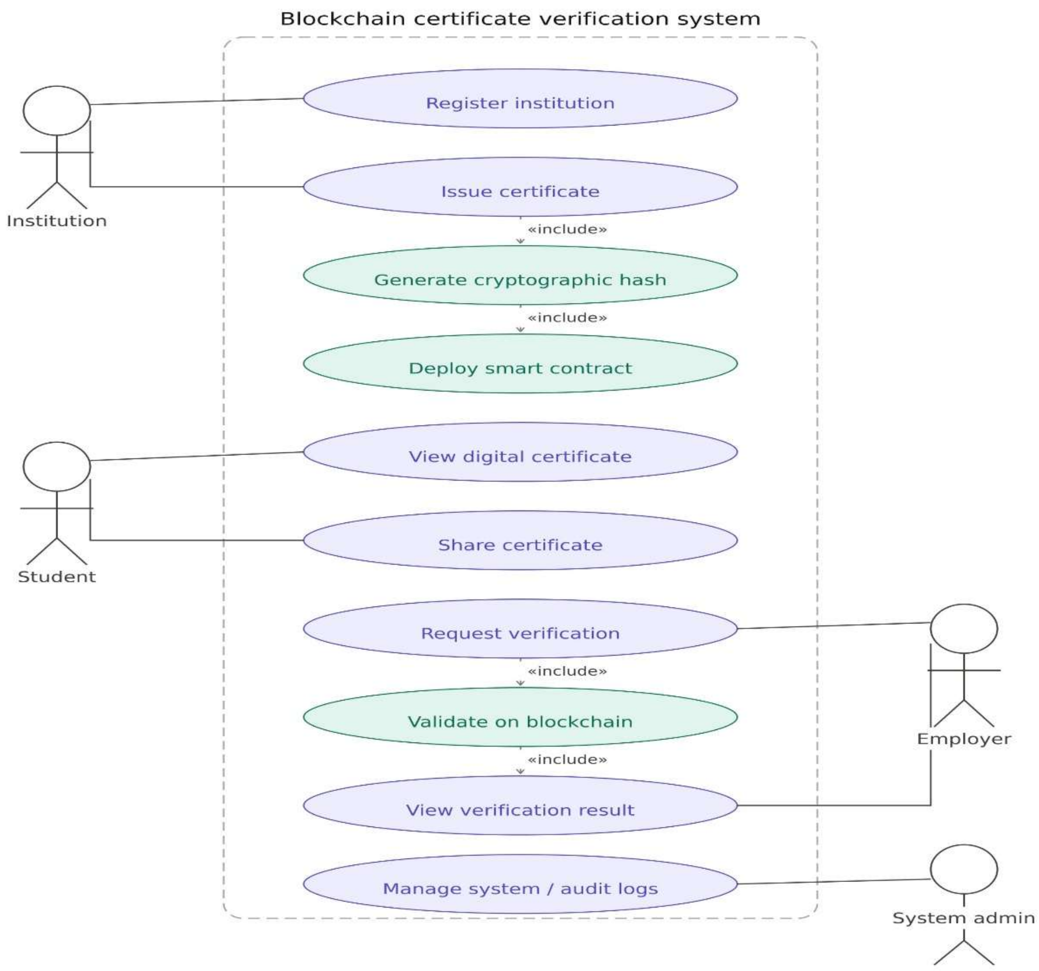

Figure-5.4.1

CHAPTER 6

PROJECT OUTCOME

6.1 System Features

- **Secure User Authentication:** Ensures only authorized users — including students, institutions, and verifiers — can access the system, protecting certificate records and blockchain transaction data from unauthorized access.
- **Certificate Issuance and Student Profiles:** Stores student academic records, issued certificates, and verification history to build reliable digital profiles, enabling institutions to manage and track credentials over time.
- **Blockchain Certificate Verification and Integrity Scoring:** Automatically validates academic certificates by cross-checking issued credential hashes against the blockchain ledger, detecting tampered or fraudulent documents and assigning an authenticity confidence score to each certificate.
- **Audit Trails and Verification Logs:** Generates detailed reports showing certificate issuance events, third-party verification requests, flagged anomalies, and overall system activity, helping administrators monitor certificate integrity and institutional compliance.

6.2 User Interface

- **Overview:** Intuitive, easy-to-use interface with clear navigation and minimal learning curve, designed for students, academic institutions, and employers verifying credentials.
- **Navigation and Layout:** Dashboards, menus, and buttons are organized for quick access to certificate issuance, verification status, and blockchain transaction records.

Dashboard:

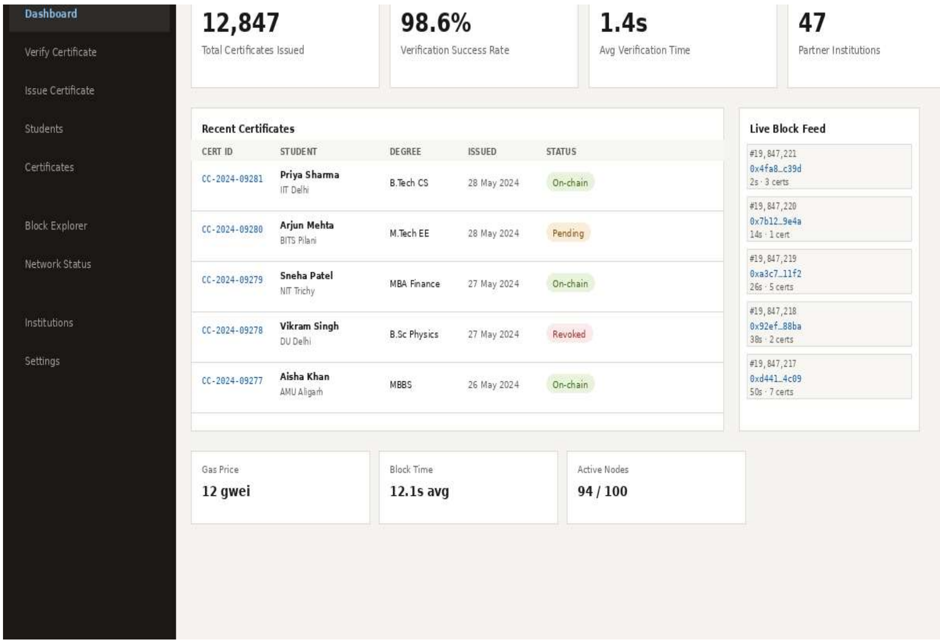

Fig 6.2.1

Verify Certificate:

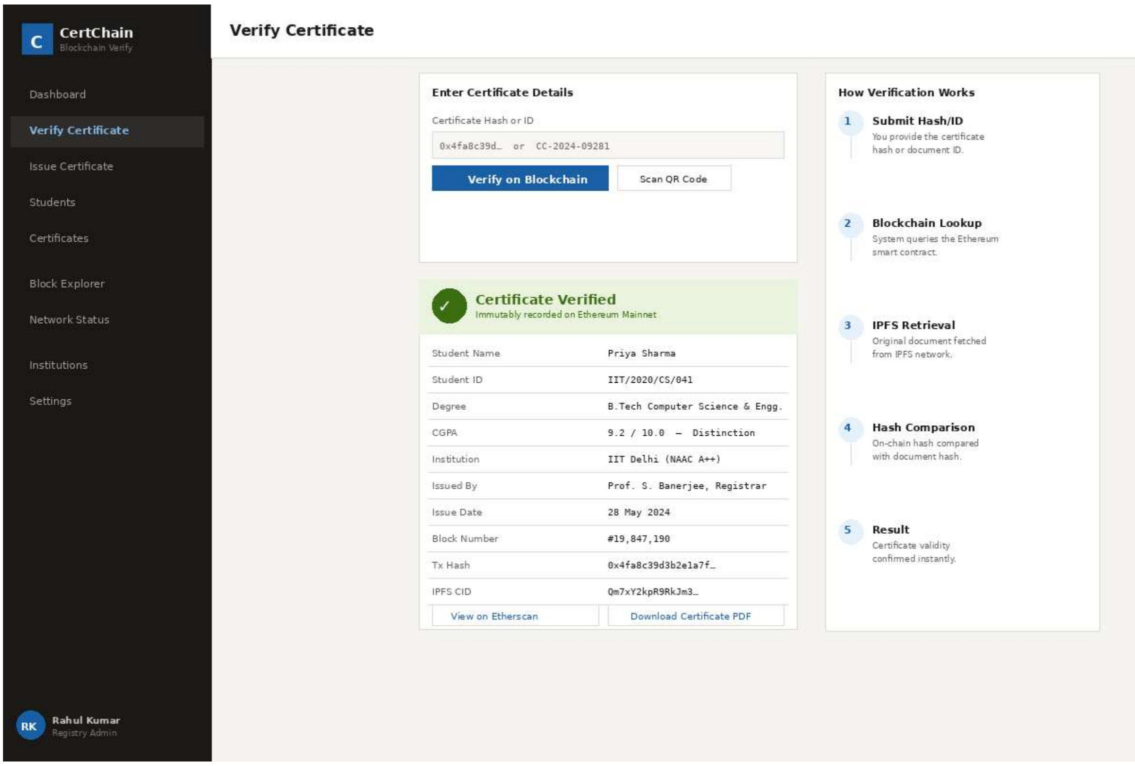

Fig 6.2.2

Issue Certificate:

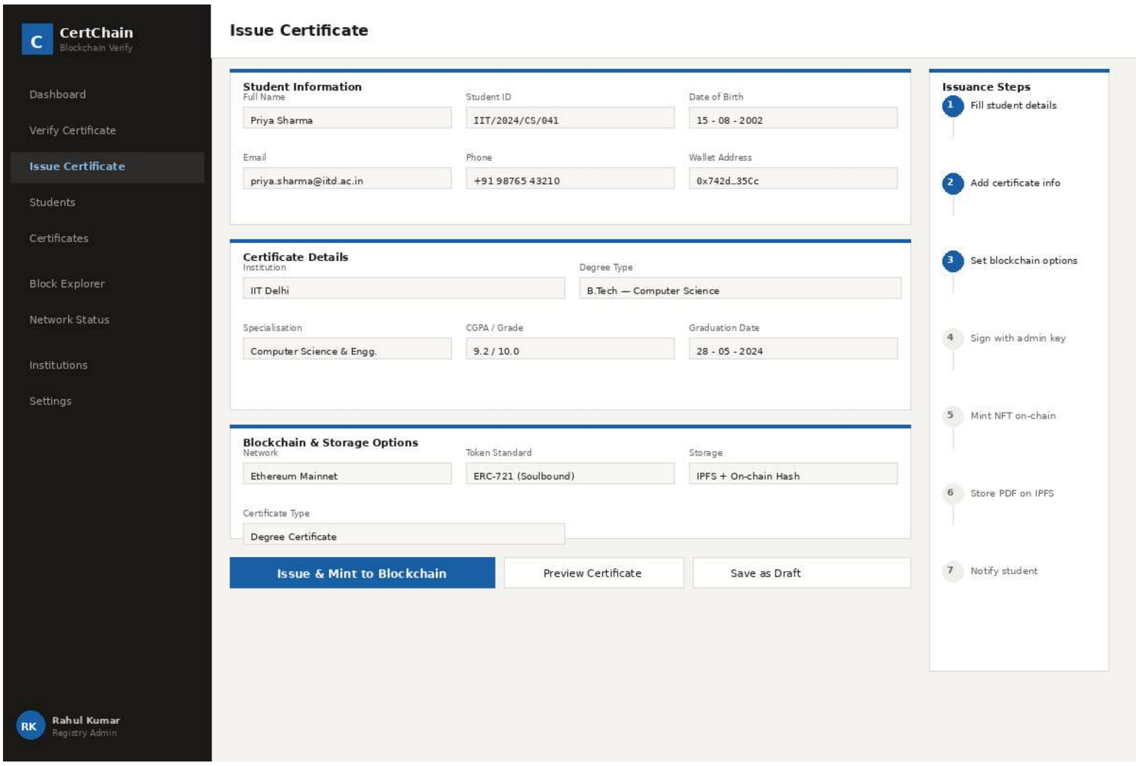

Fig 6.2.3

Students:

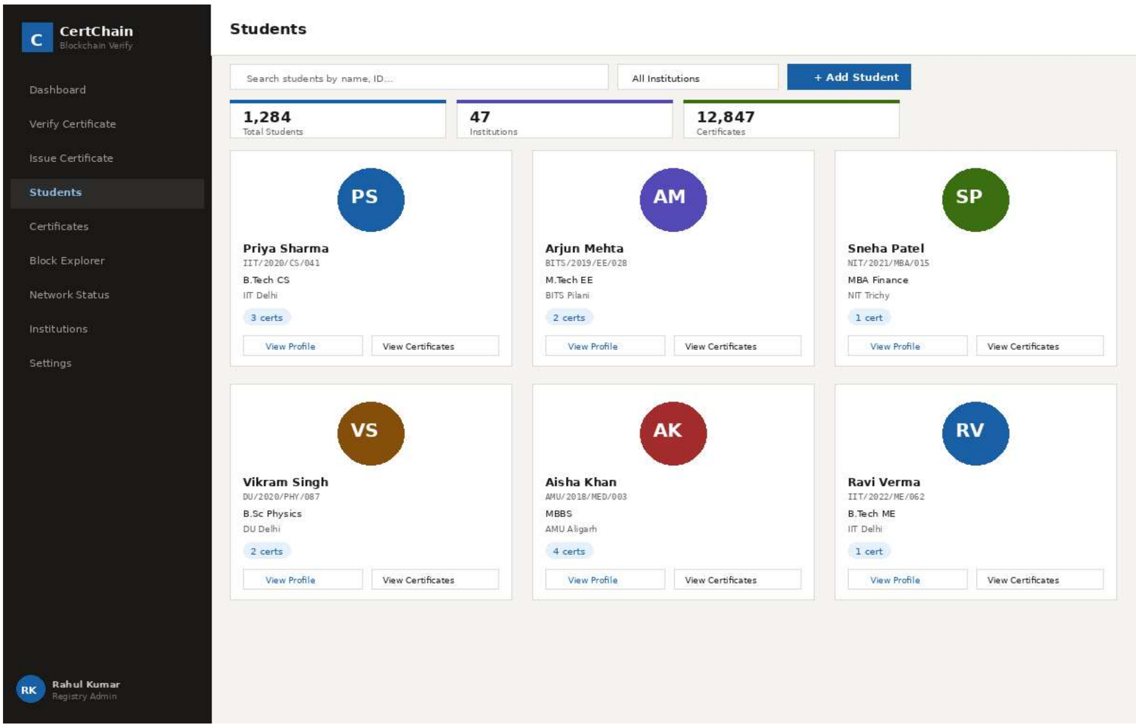

Fig 6.2.4

Certificates:

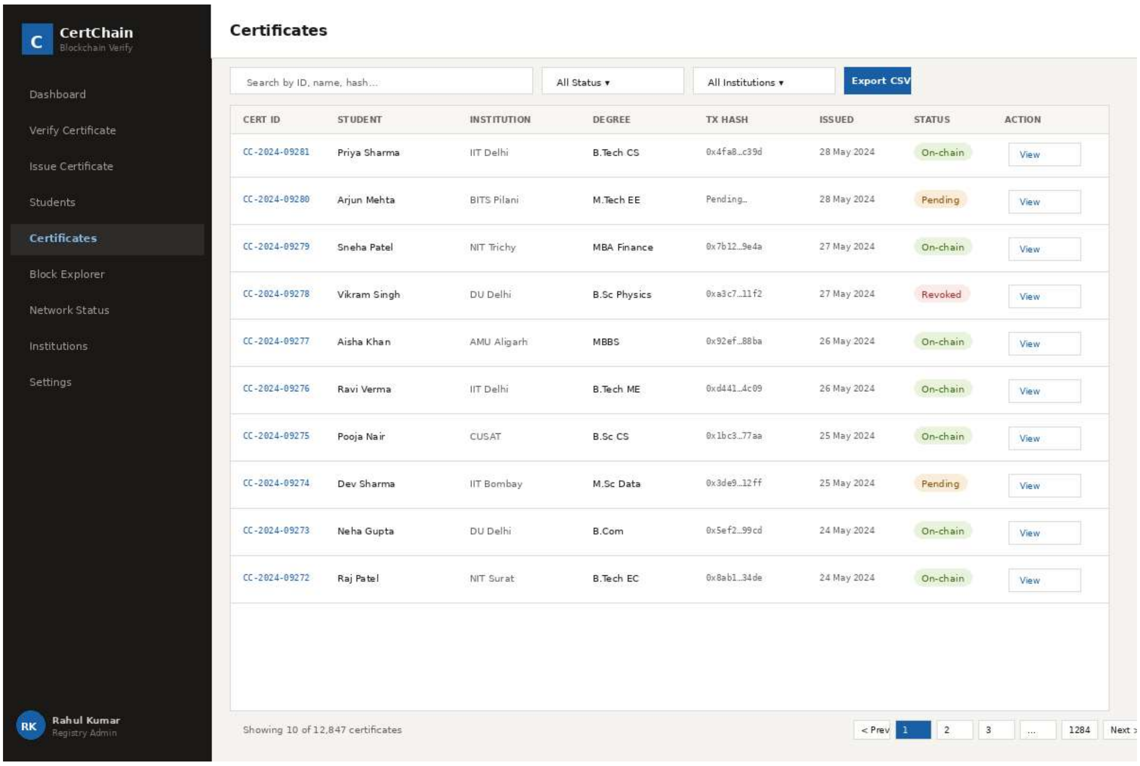

Fig 6.2.5

Institutions:

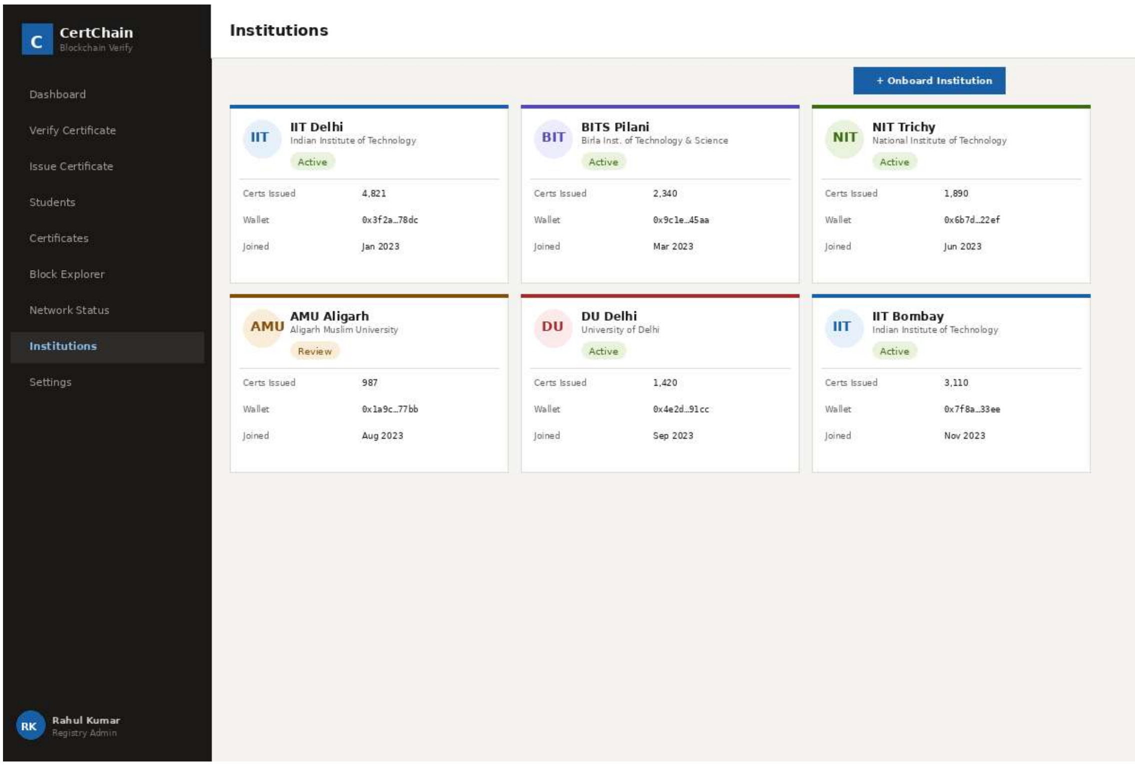

Fig 6.2.6

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